

Fusion Deep Learning for Predicting Conductivity in Electron-Doped Organic Polymers

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Cite This: *J. Am. Chem. Soc.* 2025, 147, 36233–36243

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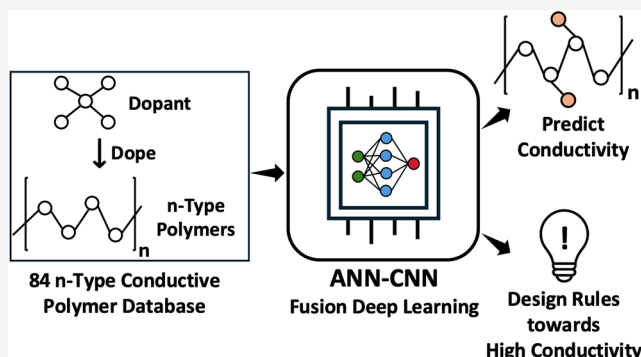


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ABSTRACT: The development of efficient and air-stable n-type organic semiconductors suitable for molecular n-doping is critical for advancing high-performance, durable organic electronic devices, including transistors, thermoelectrics, and photovoltaics. Machine learning offers a powerful approach to uncovering hidden relationships between molecular structures and their electronic properties, thereby accelerating the discovery and design of promising materials. To support this effort, we constructed a curated database comprising 84 n-type conductive polymers, each characterized by experimental measurements under n-doping conditions with 4-(1,3-dimethyl-2,3-dihydro-1H-benzimidazol-2-yl)phenyl dimethylamine (N-DMBI-H), and augmented with density functional theory calculations to provide complementary molecular descriptors. After constructing the database, we developed a fusion deep learning model that integrates convolutional neural networks with fully connected artificial neural networks to capture both structural and property-based features of the polymers. The model was trained on the data set and evaluated using leave-one-out cross-validation. The model was further applied to a test set of n-type polymers bearing oligoethylene glycol (OEG) side chains, enabling the identification of key physical factors that influence their conductivity when doped with N-DMBI-H. Finally, a double-blind experiment was conducted to validate the model's practical utility by predicting the conductivity of four BDPPV-type polymers and one N2200-type polymer doped with N-DMBI-H. For the N2200-type polymer and two of the BDPPV-type polymers, the predicted conductivities agreed with experimental values within the same order of magnitude. These results demonstrate the fusion model's reliability and establish a strong foundation for data-driven property prediction and the design of high-conductivity n-type polymers.



1. INTRODUCTION

Molecular doping of conjugated polymers is one of the most effective strategies to dramatically enhance their electrical conductivity by generating a high density of mobile charge carriers (e.g., polarons) through redox reactions between the polymer and the dopant.^{1–7} Depending on the relative redox potentials (or energy levels) of the conjugated polymer and the molecular dopant, either p-type doping (i.e., oxidation of the polymer via electron loss) or n-type doping (i.e., reduction of the polymer via electron gain) can occur.

While significant progress has been made in p-type doping over the past few decades, with reported electrical conductivities (σ) reaching as high as 2×10^5 S/cm,⁸ n-type doping remains considerably more challenging. Typical σ values for n-doped conjugated polymers are much lower, often less than 10 S/cm,^{9–15} with only one notable example where σ surpassed 2000 S/cm.¹⁶

The traditional approach to addressing materials challenges is best characterized as a feedback loop that proceeds from conceptual formulation, to material synthesis, to testing and characterization. The resulting structure–property relationships are then used to refine or reformulate the original concept,

initiating a new iteration of the process. While this method has proven effective, it is inherently slow, even when supported by substantial resource investment. In recent years, the rapid advancement of machine learning (ML) has offered a promising path to accelerate the discovery of novel materials.^{17–22}

In the field of organic electronics, ML has shown remarkable success in predicting material properties and related device performance. For example, Pyzer-Knapp et al. trained an artificial neural network (ANN) to predict frontier molecular orbital energies using data from the Harvard Clean Energy Project (CEP), a library of molecular structures generated from 26 basic building blocks.²³ Similarly, Nagasawa et al. used a convolutional neural network (CNN) and CEP data to predict

Received: May 30, 2025

Revised: September 17, 2025

Accepted: September 18, 2025

Published: September 26, 2025



power conversion efficiency of organic solar cells with over 91% accuracy.²⁴

While machine learning has been applied to doped conjugated polymers, its use has been largely limited to p-type systems. For example, Yoon et al. developed an ML model using absorbance spectra to classify p-type polymers with 100% accuracy and predict high conductivities with an R^2 of 0.984.²⁵ Jeong et al. used Bayesian-optimized tree models to tune processing conditions for p-doped thermoelectric films, maximizing power factor with minimal trial-and-error experimentation across a broad parameter space.²⁶ Sahu et al. trained SVM and GPR models on 398 p-doped polymer–dopant pairs to predict conductivity and screened over 800,000 candidates, identifying 500 for experimental validation.²⁷

To the best of our knowledge, there have been no prior reports applying machine learning to predict the properties of n-type doped conjugated polymers; we thus set out to leverage ML to accelerate the identification of conjugated polymers with appreciable electrical conductivity (σ) upon n-type doping. Our goal is to overcome the longstanding challenges associated with n-type doping by developing predictive tools that can guide molecular design before synthesis.

In this work, we present several key advances toward that goal:

- We constructed a curated database of conjugated polymers n-doped with N-DMBI-H, a widely used molecular dopant.^{2,11}
- We developed a fusion deep learning model that integrates a convolutional neural network (CNN) for capturing structural patterns and an artificial neural network (ANN) for leveraging computed quantum descriptors.
- We performed a rigorous double-blind validation to assess model generalizability.
- The resulting model demonstrates high predictive accuracy, enabling the reliable identification of high-conductivity n-type polymers before experimental testing.

These results establish a new data-driven framework for accelerating the discovery of high-conductivity n-type organic materials.

2. METHODS

2.1. First-Principles Calculations. In this study, density functional theory (DFT) with periodic boundary condition calculations were employed to characterize the fundamental electronic properties of polymers, including the highest occupied molecular orbital (HOMO), the lowest unoccupied molecular orbital (LUMO), HOMO–LUMO gap, and electron affinity (EA). To expedite the calculations, several simplification rules were implemented. For polymers with side chains containing a carbon chain longer than three carbon atoms, the long carbon chain portion will be substituted with a methyl group. For polymers whose side chains have a repeating segment, a single repeating unit will be employed in the calculations. In all remaining cases, the complete structure will be used for the computation. All geometric optimizations and property calculations were performed using Gaussian 16 with the HSE06 functional and the 6–31+G(d) basis set under periodic boundary conditions.³⁶ The theoretical properties were directly extracted from the Gaussian output files.

2.2. Structural Coulomb Matrix. Among various molecular representations, the Coulomb matrix offers a simple yet effective global descriptor that captures the three-dimensional structure of a molecule by mimicking the electrostatic interactions between its nuclei. The elements of the Coulomb matrix are computed using the following equation

$$V_{ij} = \begin{cases} 0.5Z_i^{2.4}, & \text{if } i = j \\ \frac{Z_i Z_j}{|R_i - R_j|}, & \text{if } i \neq j \end{cases} \quad (1)$$

where Z_i is the atomic number of the i th atom, R_i is its position vector, and $|R_i - R_j|$ is the Euclidean distance between atoms i and j . The diagonal elements reflect the interaction of an atom with itself, expressed as a polynomial fit of the atomic energy to the nuclear charge Z_i . In contrast, the off-diagonal elements represent the Coulomb repulsion between nuclei i and j .

To make the size of the Coulomb matrix compatible with the CNN input, all Coulomb matrices are zero-padded to the size of 150×150 . Although such padding could potentially influence the performance of the CNN, it is a necessary trade-off to maintain the consistency of the inputs.

The Coulomb matrix is symmetric and invariant to translation and rotation. However, permutation invariance cannot be achieved with the Coulomb matrix, as its values change when the order of two atoms is switched. To address this issue, a norm-2 sort was applied. The norm-2 sort is defined by the following equation

$$\|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2} \quad (2)$$

where $\|x\|_2$ denotes the Euclidean norm of a vector $x \in \mathbb{R}^n$, and x_i represents the i th component of x .

In the sorted Coulomb matrix, atoms with higher atomic numbers are assigned higher positions. For atoms sharing the same atomic number, the atom experiencing the most substantial Coulomb repulsion from neighboring atoms is given a higher position. This sorting approach achieves permutation invariance for the input. These matrices were generated using the Python packages DeepChem and RDKit.

2.3. Machine Learning. Artificial neural networks (ANNs) and convolutional neural networks (CNNs) are computational models inspired by the structure and functioning of biological neural networks. ANNs consist of interconnected layers of artificial neurons, where each neuron processes inputs and transmits the results to subsequent layers. They are widely employed for tasks such as classification, regression, and clustering.

In contrast, CNNs are specialized deep learning architectures designed to process data with a grid-like topology, such as images. These networks have achieved significant success in tasks including image classification, object detection, and segmentation. In this study, CNNs were utilized to uncover hidden relationships between polymer structures and their conductivity.

Building on the strengths of both ANN and CNN, we developed a fusion deep learning model that integrates these two architectures. As shown in Figure 5, numerical data were input into the ANN component, while structural data were processed by the CNN component. A fully connected layer combined the outputs from both components, enabling the model to predict polymer conductivity.

For hyperparameter optimization, we employed the random search, an efficient technique to explore high-dimensional hyperparameter spaces. Compared to grid search, random search increases the likelihood of identifying optimal hyperparameters, thereby enhancing the model performance.

Our fusion deep learning model comprises three hidden layers in both the ANN and CNN components. Given the relatively small size of the data set (84 cases), adding more hidden layers was avoided to prevent overfitting. Detailed descriptions of each layer and their configurations are provided in Table 1.

2.4. Electron-Doped Polymer Database. In the realm of n-type doped organic semiconductors (OSCs), no existing conducting polymer database is available for predicting experimental conductivity. To address this gap, we developed a specialized database that integrates experimental conductivity data collected from the literature, density functional theory (DFT)-computed electronic descriptors, and DFT-

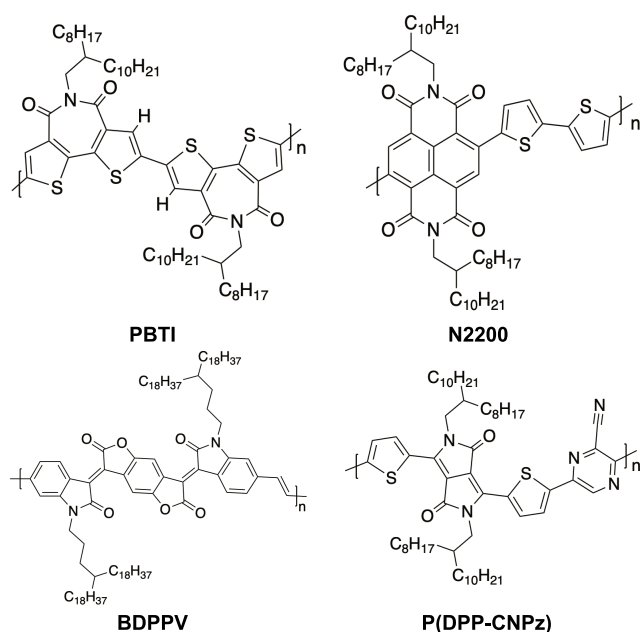
Table 1. Hyperparameters of the Fusion Deep Learning Model

component	hyperparameter	value
CNN	Conv Filters (Layer 1)	128
	Conv Kernel Size (Layer 1)	3
	Conv Filters (Layer 2)	32
	Conv Kernel Size (Layer 2)	3
	Conv Filters (Layer 3)	128
	Conv Kernel Size (Layer 3)	3
	Dropout (Conv Layer 3)	0.2
ANN	Dense Units (CNN Branch)	128
	Dense Units (Features Branch 1)	128
	Dropout (Features Branch 1)	0.2
	Dense Units (Features Branch 2)	96
	Dropout (Features Branch 2)	0.3
	Dense Units (Features Branch 3)	192
	Dropout (Features Branch 3)	0.1
combined	Dense Units (Combined Branch)	192
	Dropout (Combined Branch)	0.4
general	Learning Rate	0.00033

optimized polymer structures. This database facilitates more accurate and efficient predictions of OSC properties.

The database comprises 84 experimentally tested polymers that can be doped with N-DMBI-H to achieve good conductivity. These polymers were carefully selected from the literature, focusing on four seed polymers: poly(2,2'-bithiophene-3,3'-dicarboximide) (PBTI), benzodifurandione-based poly(phenylene vinylene) (BDPPV), and poly[*N,N'*-bis(2-octyldodecyl)-naphthalene-1,4,5,8-bis-(dicarboximide)-2,6-diyl]-*alt*-5,5'-(2,2'-bithiophene) (N2200), poly-[(diketo-pyrrolopyrrole)-*alt*-(pyrazine-2-carbonitrile)] P(DPP-CNPz), as shown in Figure 1.

To ensure the model's general applicability, the database also includes polymers with backbones different from the four seed polymers. When a specific polymer was reported multiple times in the literature, the instance with the highest conductivity was selected for inclusion, ensuring that the database represents the best experimental performance for each polymer.

**Figure 1.** The four conjugated backbones and their corresponding structures.

3. RESULTS AND DISCUSSION

3.1. Training Database. The success of the data-driven method relies heavily on the quality of the training data set. Among n-dopants, 1,3-Dimethyl-2-phenyl-2,3-dihydro-1*H*-benzimidazole (DMBI-H) derivatives, particularly (4-(1,3-dimethyl-2,3-dihydro-1*H*-benzimidazol-2-yl) phenyl) dimethylamine (N-DMBI-H), are considered the most popular and successful air-stable n-dopants due to their excellent stability and doping activity.¹³ N-DMBI-H has been applied to dope a wide range of semiconductors, including both inorganic and organic materials.^{37–39} Table 2 highlights recent applications of N-DMBI-H in the organic semiconductor field.

Table 2. Thermoelectric Properties of Polymers Doped with N-DMBI-H, Including LUMO Energy Levels, Electrical Conductivity (σ), and Maximum Seebeck Coefficient ($S_{\max}$)

polymers	LUMO (eV)	σ (S/cm)	$S_{\max}$ ($\mu\text{V}\cdot\text{K}^{-1}$)	refs
PBTI	−3.48	2.0×10^{-3}	-7.8×10^2	13
BDPPV	−4.01	2.6×10^{-1}	-3.0×10^2	28
FBDPPV	−4.17	1.4×10^1	-1.4×10^2	28
LPPV-1	−4.49	1.1×10^0	-1.7×10^2	29
PDPF	−4.11	1.3×10^0	-2.4×10^2	30
PFCITVT	−4.03	3.8×10^1	-2.6×10^2	31
PNDI2OD-2T(N2200)	−3.75	1.2×10^{-3}	—	32
f-BTI2TEG-FT	−3.82	1.0×10^2	8.2×10^1	33
PNDTI-BBT-DP	−4.40	5.0×10^0	-1.7×10^2	34
P(FBDOPV-2T-C12)	−4.00	4.2×10^{-2}	-2.7×10^2	35

To support our study, we curated a database of 84 experimentally tested n-type polymers that can be doped with N-DMBI-H, compiled from the literature (see the Supporting Information). The experimental database comprises conductivity data for these polymers. The dopant concentration and other process conditions could strongly influence the observed conductivities. To ensure data integrity, we applied strict criteria for conductivity data collection, including consistency in dopant use, dopant molar ratio, and measurement conditions (see the Supporting Information for more details). We excluded conductivity data that used catalysts, such as gold nanoparticles,⁴⁰ which are known to accelerate electron transfer and artificially inflate conductivity. Because experimental measurements on electron-doped n-type polymers are relatively scarce, the database size is inevitably limited. However, this careful curation ensures consistency and reliability, making it a solid foundation for model development. In the Supporting Information, Figure S1 shows the distribution of log-(conductivity) for the training set, and Table S2 summarizes the statistics for each polymer family.

To visualize the database, the monomers of each polymer were clustered by combining Morgan fingerprints⁴¹ and the t-distributed stochastic neighbor embedding (t-SNE) method,⁴² as shown in Figure 2. The t-SNE algorithm reduces dimensionality while preserving local structures. It identifies clusters of similar molecules and maps these to neighboring points in two-dimensional space.

Here, all 84 monomers are divided into five categories: BDPPV-type (blue circles), DPP-type (yellow downward triangles), N2200-type (bluish green diamonds), PBTI-type (sky blue triangles), and others (orange squares). The regions are almost clearly separated from each other, demonstrating both the power of the t-SNE algorithm and the computer's

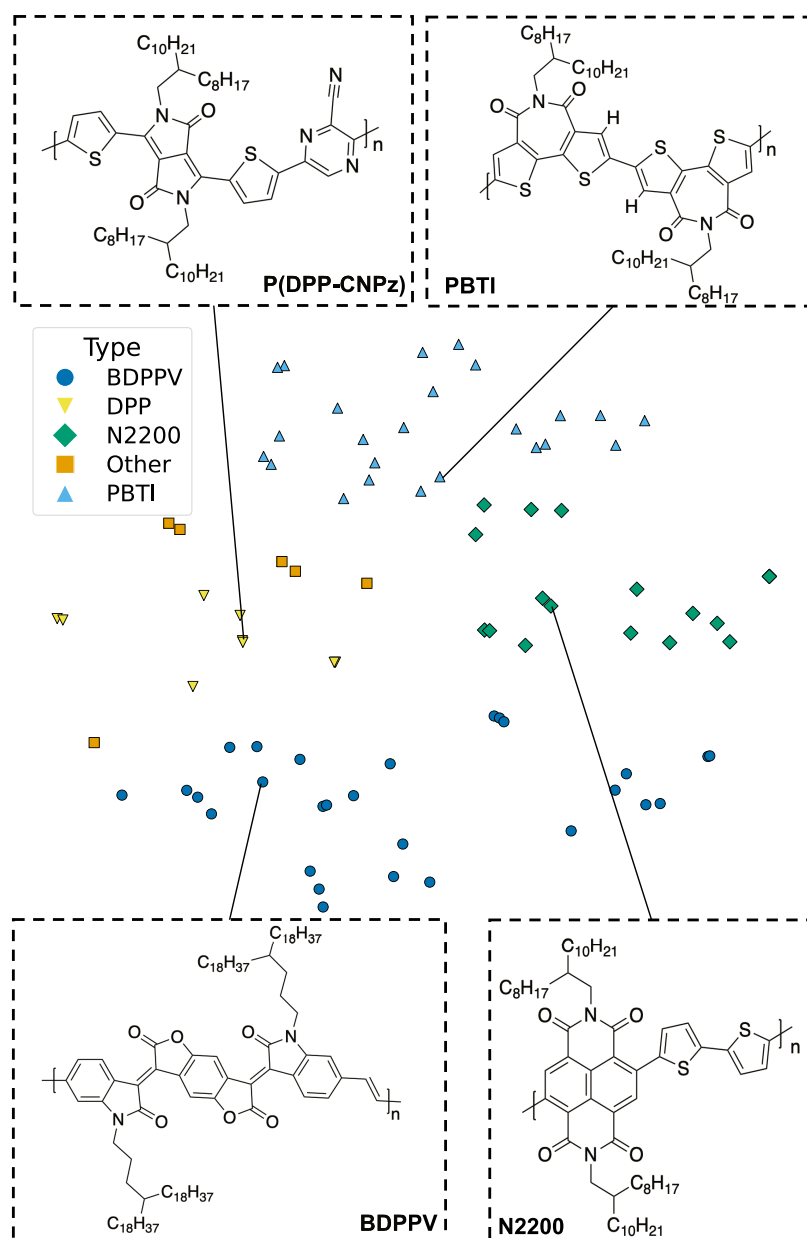


Figure 2. Visualization of the n-type doping database generated using a t-SNE clustering algorithm, where each data point represents an individual polymer. In the plot, polymers are distinguished by both color and marker shape: BDPPV-type polymers are depicted as blue circles, DPP-type polymers as yellow downward triangles, N2200-type polymers as bluish green diamonds, other polymers as orange squares, PBTI-type polymers as sky blue triangles.

ability to distinguish between structures of different monomer types. This t-SNE visualization is used solely to illustrate structural diversity and is not involved in the subsequent machine learning workflow.

To represent the polymer structures, the Coulomb matrix (CM) was used as a global descriptor.⁴³ This simple yet effective method approximates the electrostatic interactions between nuclei and captures the three-dimensional structure of the monomers. Unfortunately, the CM does not exhibit permutation invariance for the input; any change in the atom order produces a distinct matrix. To address this issue, a norm-2 sorted Coulomb matrix (SCM) was employed (see the Section 2 for details).⁴⁴ An example of the SCM is presented in Figure S2 in the Supporting Information. The SCM achieves invariance with respect to translation, rotation, and permutation, making it an

ideal input for chemical structural machine learning algorithms. Moreover, the norm-2 SCMs contain more chemical information than the regular CM. Specifically, atoms with higher atomic numbers are assigned higher positions, and the atom experiencing the strongest Coulomb repulsion from its neighbors is similarly positioned at a higher level.

For n-type doping, the highest occupied molecular orbital (HOMO) of the dopant must be higher than or close to the OSC's lowest unoccupied molecular orbital (LUMO), enabling electrons to transfer from the dopant to the polymer. In contrast, p-type doping involves the transfer of electrons from the OSC to the dopant, which requires the HOMO of the OSC to be higher than or close to the LUMO of the dopant. This process effectively alters the charge carrier concentration within the polymer, leading to improved electrical conductivity. Electron

affinity (EA) is another critical factor in determining the efficiency of chemical doping.⁴⁵ Polymers with higher EA values exhibit a greater tendency to accept electrons, enhancing the stability of the doped state and improving overall conductivity.

To enable effective ML training and predictive modeling, the electron-doped n-type polymer data set was augmented with first-principles data as key descriptors. These include optimized polymer geometries for constructing structural Coulomb matrices, as well as crystalline HOMO and LUMO energies, the HOMO–LUMO energy gap, and monomer electron affinity (see the Section 2 for details). Notably, the energy gap is defined as the difference between crystalline LUMO and HOMO. While this may seem redundant given the availability of HOMO and LUMO values, it helps reduce noise and provides the model with an explicit interaction it would otherwise need to infer. Furthermore, the energy gap is a chemically meaningful descriptor, often correlating with electrical conductivity. To complement the information from the crystalline LUMO, we also include the monomer EA, calculated as the energy difference between a fully optimized monomer and its anion. These parameters represent essential electronic structure features that govern charge transfer in n-doped polymers.

3.2. Statistical Descriptors and Correlation. Key statistical descriptors were employed to characterize the distribution of conductivity values. Simultaneously, correlations between electronic properties and conductivities were quantified using a correlation heatmap and scatter plots for each descriptor against conductivity.

Figure 3A illustrates the distribution of conductivity values, revealing a skewed data set with 71 data points concentrated in the 0 to 10 range. Notably, the conductivity values span from a minimum of 0.0000020 S/cm to a maximum of 143.00 S/cm, with a mean of 9.47 S/cm, a median of 0.19 S/cm, and a standard deviation of 25.00 S/cm. ML models trained on such an imbalanced distribution are likely to inaccurately predict higher values while skew toward the more abundant lower values. To address this issue, a logarithmic transformation of the conductivity data was employed, as shown in Figure 3B. The transformed data range from a minimum of -5.82 to a maximum of 2.16 , with a mean of -1.10 , a median of -0.72 , and a standard deviation of 1.97 . This logarithmic representation offers a more balanced distribution, providing a robust foundation for subsequent analysis and model training.

Figure 4 illustrates the fundamental correlations between the electronic descriptors and conductivities. Although the correlations between logarithmic conductivity and crystalline LUMO or monomer EA are small, they are still notably stronger than those between logarithmic conductivity and either the HOMO or the gap. These results are consistent with the n-type doping mechanism, in which electrons are introduced into the polymer's LUMO,⁴⁶ affirming the validity and appropriateness of the selected descriptors. The correlation between crystalline LUMO and monomer EA is -0.76 , suggesting that the property information provided by these two descriptors is not fully redundant and that each contributes partially distinct information to the model. Scatter plots for all the electronic descriptors against logarithmic conductivity are provided in the Supporting Information.

3.3. An ANN-CNN Fusion Deep Learning Model. To use a data-driven approach to predict polymer conductivity, various ML models, including ANN, CNN, and a CNN-ANN fusion model, were implemented and tested in this work. Due to the limited size of the database, all three models were trained on the

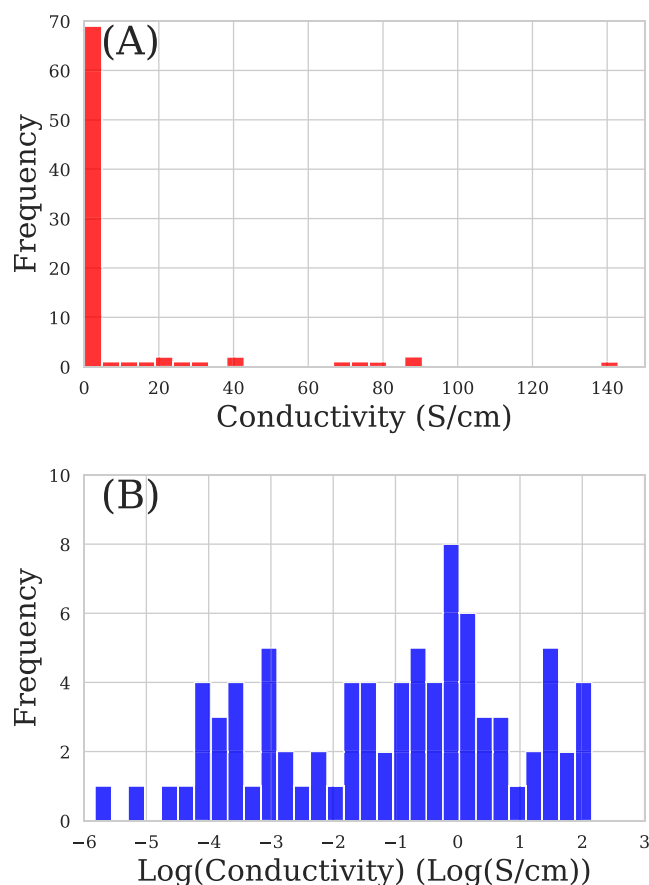


Figure 3. (A) Distribution of conductivity data with a bin width of 5, revealing a pronounced skew. The x-axis represents the conductivity of polymers doped with N-DMBI-H at a 20% molar ratio, while the y-axis shows the number of samples (frequency) in each bin. (B) Distribution of logarithmically transformed conductivity data, which is more balanced compared to the original conductivity distribution.

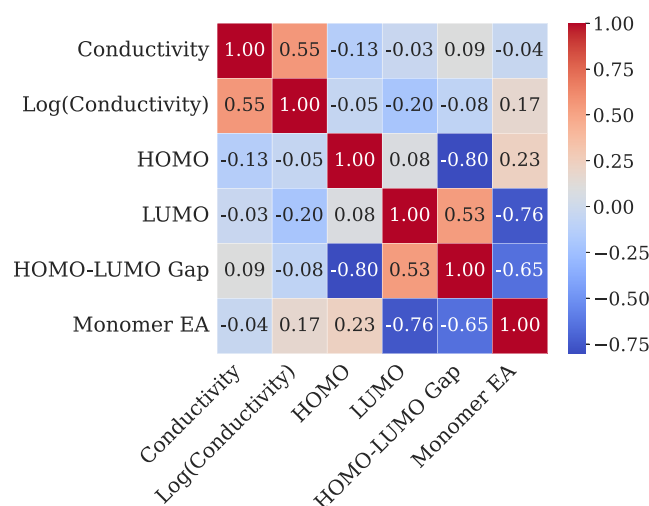


Figure 4. Pearson correlation heatmap for electronic descriptors versus conductivity. A color scale bar on the right maps the Pearson correlation coefficient, with red indicating a positive correlation and blue indicating a negative correlation.

complete database of 84 polymers, with hyperparameters initialized using random search. To validate the capability and effectiveness of all the models, cross-validation techniques,

particularly leave-one-out cross-validation (LOOCV), were applied. The performance of each model was evaluated using mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE) metrics (see the Section 2 for details and the Supporting Information for parity plots of those three models).

LOOCV is a cross-validation technique in which each data point is used as a test set once while the remaining data points form the training set, with this process repeated for every data point. For a data set with 84 entries, LOOCV generates 84 test results, each contributing to the statistical distribution of the performance for each ML model. This approach is particularly useful here since the database is relatively small, allowing for a thorough evaluation of model performance by utilizing all available data for both training and testing. As expected, the performance of the ML models varied, as shown in Table 3, reflecting their different architectures and underlying mechanisms.

Table 3. LOOCV Performance Metrics

model	MSE	RMSE	MAE
ANN	4.96	2.23	1.72
CNN	4.77	2.18	1.76
Fusion Model	4.67	2.16	1.76

The ANN model yielded an MSE of $4.96 \log^2(\text{S/cm})$ and an MAE of $1.72 \log(\text{S/cm})$, indicating moderate predictive accuracy. In comparison, the CNN model, specifically designed to process grid-like data, such as the SCM, achieved an MSE of $4.77 \log^2(\text{S/cm})$ and an MAE of $1.76 \log(\text{S/cm})$. The CNN's lower MSE and RMSE compared to the ANN indicate that the CNN model exhibits fewer large deviations. The lower MSE and RMSE observed in the CNN model suggests that its predictions capture the broad conductivity range arising from variations in polymer structure. The low MAE in the ANN model indicates that the calculated electronic descriptors are particularly critical for accurately predicting the conductivity of certain polymers. This observation motivates the use of structural data to establish a baseline prediction, with the ANN model subsequently employed to calibrate and refine the results.

We thus developed a fusion deep learning model that leverages the strengths of both CNN and ANN, as illustrated in Figure 5. This model was trained to capture the intricate relationship between polymer structures and their properties. During training, the CNN extracted structural patterns from the sorted Coulomb matrix, while the ANN analyzed computed material properties. These networks were integrated via a fully

connected layer, enabling the model to synthesize both structural and property-based information.

Compared to the standalone ANN and CNN models, this fusion model demonstrated strong predictive performance, achieving an MSE of $4.67 \log^2(\text{S/cm})$. This result reveals that conductivity is not solely determined by a single descriptor but is influenced by both the properties and structures of the polymers. With a better performance than either standalone model, the fusion deep learning model enhances pattern recognition by leveraging correlations between structural and property data, leading to more robust and generalizable predictions.

It is important to note that, due to the limited size of the database, the fusion deep learning model's predictive power varies across different polymer types, as evidenced by the LOOCV results in Table 3 (see the Supporting Information for details of the LOOCV results). Fourteen LOOCV tests yielded conductivity values within the same order of magnitude as the experimental data. Among these, six correspond to the N2200-type, five to the BDPPV-type, two to the PBTI-type, and one to the DPP-type. These results suggest that the fusion model performs considerably better for the N2200-type and BDPPV-type than for other structures. Furthermore, an analysis of the logarithmic MSE values revealed that 25 models achieved an MSE below 1, while 57 models recorded an MSE lower than the overall average. This suggests that over half of the tests perform above average.

The fusion deep learning model was further validated using a 5-fold cross-validation within the $\pm 15\%$ hyperparameter neighborhood and a 90/10 training/validation split, with the results provided in the Supporting Information.

After validating the capability and effectiveness of the fusion deep learning model, it will subsequently be employed for double-blind experimental validation and for predicting the conductivity of new polymers.

3.4. Double-Blind Experimental Validation. We conducted a double-blind experiment to evaluate the practical application of the fusion deep learning model. The experimental group synthesized five n-type polymers, four of the BDPPV-type and one of the N2200-type, which were doped with N-DMBI-H, followed by conductivity measurements. Structures of these polymers are presented in Figure 6. Subsequently, DFT calculations and geometric optimizations for the simplified structures (see the Section 2 for details) were performed to extract the relevant electronic descriptors for prediction. The electronic descriptors of these five polymers are shown in Table 4.

With the structural data and descriptors in hand, we applied the fusion deep learning model to predict the conductivity of these five polymers when doped with 20% N-DMBI-H. The experimental and ML-predicted conductivities are summarized in Table 5. More detailed experimental results are provided in the Supporting Information.

When comparing the experimental and predicted results, some predictions closely match the experimental values, while others show notable discrepancies. In particular, the predictions for three polymers, N2200-OEG, BDOPV-TVT, and FBDOPV-TVTCN, aligned very well with the experimental data. For these cases, the predicted conductivity not only matched the order of magnitude of the experimental conductivity but also was quantitatively close, with the predicted logarithmic conductivity within 15% of the experimental value.

As for the other two polymers, BDOPV-TVTCN and FBDOPV-TVT, they exhibited noticeable prediction errors,

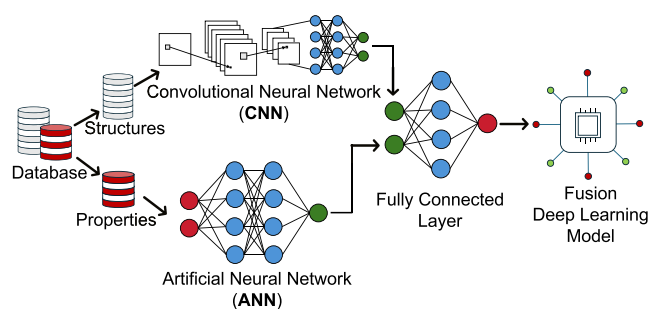


Figure 5. Fusion deep learning architecture combines ANN and CNN.

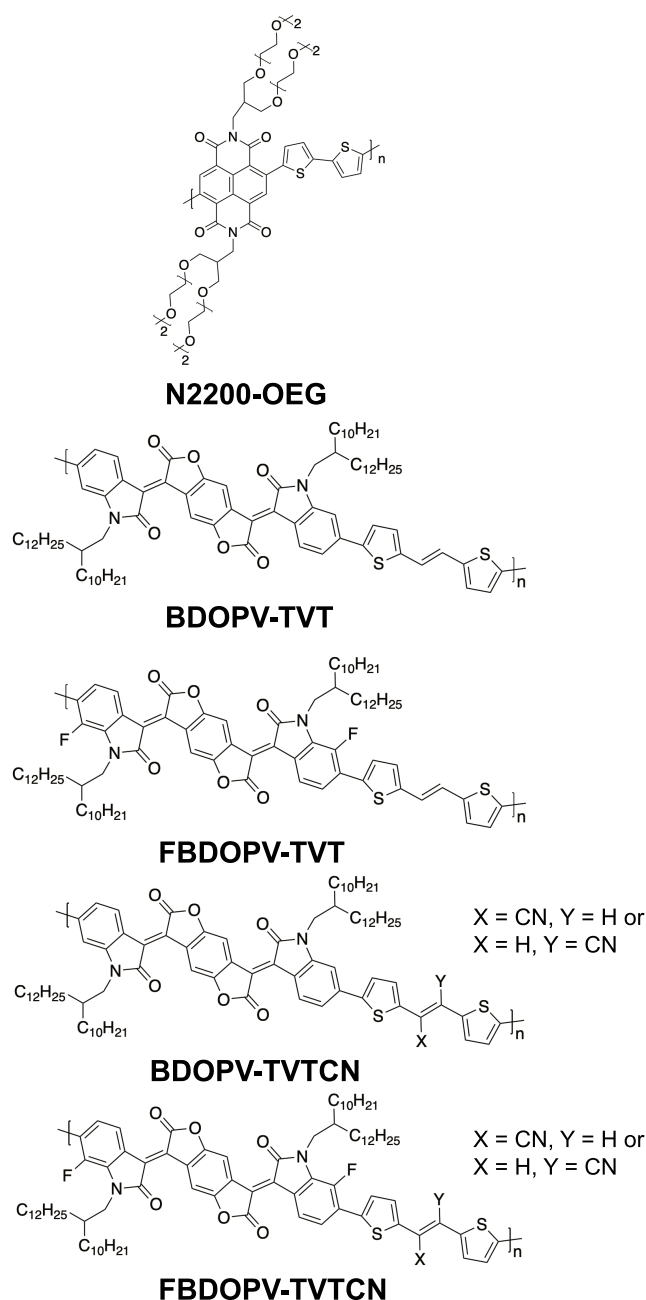


Figure 6. Experimental structures of five new polymers that served as the test set.

Table 4. Electronic Properties of Five New Polymers in eV

polymer	HOMO	LUMO	gap	EA
N2200-OEG	−5.51	−4.17	1.33	2.53
BDOPV-TVT	−5.25	−4.10	1.15	2.78
BDOPV-TVTCN	−5.51	−4.26	1.25	2.87
FBDOPV-TVT	−5.35	−4.25	1.09	2.94
FBDOPV-TVTCN	−5.61	−4.40	1.21	3.02

which may be attributed to limitations in the training data set. In the region of $\log(\text{conductivity}) < -3$, the training set contains only 20 data points, representing about 24% of the total set. These points are distributed across five polymer families (N2200: 7; PBTI: 6; BDPPV: 4; DPP: 2; Other: 1), leaving only a few examples per family. This data scarcity, combined with structural diversity, introduces a systematic bias that largely

explains the relatively higher prediction errors observed for low-conductivity polymers (see the parity plot in Figure S9 in the Supporting Information).

Compared to the mean $\log(\text{conductivity})$ values of the training set (see the Supporting Information), the predicted value of -1.70 for the N2200 family and the mean value of -2.33 for the BDPPV family are both far from the median and mean of their respective families. This suggests that the fusion model does not simply predict central values for each family.

The agreement between experimental and predicted conductivities highlights the reliability of the fusion deep learning model in capturing broad trends. However, because two polymers in the test set are markedly mis-predicted, the model is best viewed as a rapid screening tool rather than a definitive predictor. Nevertheless, the strong overall correlation between predictions and experiments confirms the model's capacity to accelerate materials design.

3.5. Test-Set Neighbor Analysis. To elucidate the prediction mechanism of the fusion deep learning model, we analyzed the neighboring data points of the five new polymers from the double-blind experiment. As shown in Figure 7, the stars represent the new polymers, illustrating their similarity to monomers in the training set. We computed the Tanimoto distance between each new polymer and all training data points to identify the five nearest neighbors for each case.

The nearest neighbors of N2200-OEG are TEG-N2200, PNDI2TEG-2T, PNDI2C8TEG-2Tz, N2200, and PNDI2-TEG-T2DEG, all belonging to the N2200-type polymer family, with a Tanimoto-distance-weighted average logarithmic conductivity of -2.33 . Similarly, BDOPV-TVT and BDOPV-TVTCN share the same neighbors—BDOPV-TCCT, BDOPV-2T, BDOPV-T, P3a, and BDPPV—all BDPPV-type polymers, yielding a weighted average logarithmic conductivity of -2.79 . Likewise, FBDOPV-TVT and FBDOPV-TVTCN share the same neighbors: P(FBDOPV-2T-C12), PFCITVT, PDTz, FBDPPV, and UFBPPV, which are all BDPPV-type polymers, with a weighted average logarithmic conductivity of -0.85 .

As shown in Table 6, the fusion deep learning model reduces the MAE by 39% and the RMSE by 29% relative to the neighbor-average conductivity baseline for predicting the conductivity of the five new polymers. These gains indicate that the model captures intrinsic structure–property relationships, rather than merely averaging the values of neighboring data points. This finding highlights the fusion model's ability to uncover the underlying patterns linking molecular structure, electronic properties, and conductivity.

To further demonstrate the fusion deep learning model's ability to capture chemical variation, we applied several baseline models, including a simple average model, a random forest, a kernel ridge regression model, and a Bayesian-optimized ANN model, to the five new polymers. The fusion deep learning model achieved a substantially lower error than all baselines, with an RMSE of $0.84 \log(\text{S/cm})$ (see Supporting Information for performance details).

Furthermore, we applied the Deep Ensemble method to assess prediction uncertainty. A set of neural networks with the same architecture and hyperparameters as the fusion model was trained independently, each initialized with a different random seed. The ensemble was expanded to 40 models, with performance details provided in the Supporting Information. For all five new polymers, the predictive means and standard

Table 5. Experimental and Machine Learning Predicted Conductivity of Five New Polymers

polymer	experimental conductivity (S/cm)	experimental log(conductivity)	ML predicted log(conductivity)	ML predicted conductivity (S/cm)
N2200-OEG	2.00×10^{-2}	−1.70	−1.51	3.09×10^{-2}
BDOPV-TVTV	1.14×10^{-2}	−1.94	−1.94	1.15×10^{-2}
BDOPV-TVTCN	2.00×10^{-3}	−2.70	−1.86	1.38×10^{-2}
FBDOPV-TVTV	6.47×10^{-4}	−3.19	−1.53	2.95×10^{-2}
FBDOPV-TVTCN	2.99×10^{-2}	−1.52	−1.41	3.89×10^{-2}

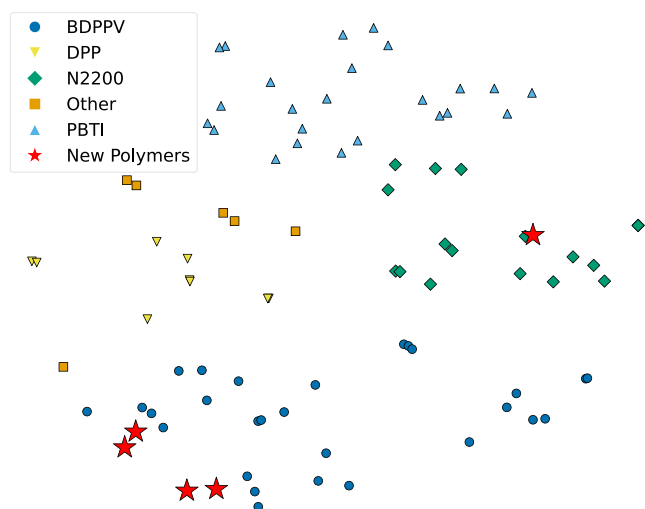


Figure 7. Visualization of the database using a t-SNE algorithm, where each data point represents an individual polymer. New polymers are indicated by red stars.

Table 6. Performance of the Fusion Deep Learning Model Versus a Tanimoto-Distance-Weighted Neighbor-Average Baseline on the Five New Polymers

metric	fusion	average baseline
MAE (log(S/cm))	0.56	0.92
RMSE (log(S/cm))	0.84	1.19

deviations remained nearly unchanged, confirming the high robustness of the fusion model.

3.6. Predicting Conductivities of Polymers. To evaluate the fusion model's capability in predicting the conductivity of unknown species, we further constructed 20 virtual polymers as inputs for the model. All the virtual polymers share the exact backbones present in the training set. Therefore, the OEG substitution is the only new structural variable. According to DFT calculations, OEG side chains can increase EAs of polymers by 0.1–0.2 eV, enhancing their tendency to gain electrons. Additionally, OEG side chains lower the LUMO energy levels compared to the original structures. Furthermore, the incorporation of OEG side chains alters the structural characteristics of the polymers, introducing additional variables that may influence their properties. For example, Kim et al. verified that the OEG side chain can serve as a solubilizing group, significantly increasing the electron mobility of n-channel polymer semiconductors and, consequently, their conductivity.⁴⁷ Similarly, Chen et al. reported that for poly((2,5-diyl-2,3,5,6-tetrahydro-3,6-dioxo-pyrrolo(3,4-c)pyrrole-1,4-diyl)-alt-(2,2':5',2''-terthiophene-5,5''-diyl)) (PDPP3T) conjugated polymer backbones, the presence of OEG side chains resulted in a reduced π – π stacking distance, higher hole mobility, a smaller

optical band gap, an increased dielectric constant, and a larger surface energy.⁴⁸

Our fusion model's predictions show that OEG side-chain substitution modulates polymer conductivity. Among the test set, seven candidates are predicted to exhibit N-DMBI-H-doped conductivities between 1.07 and 2.04 S/cm, positioning them in the 69–76% percentile range of the training distribution, about an order of magnitude above the median value. These exemplars outline the structural design space most conducive to high conductivity and thus serve as practical guidelines for synthesizing next-generation materials. Detailed outputs from the CNN, ANN, and fusion models are provided in the [Supporting Information](#).

Based on the prediction results, we identified three design rules:

- The first design rule arises from the properties of OEG-substituted PNDTI-BBT-DP, whose predicted conductivity is 2.62 S/cm when doped with N-DMBI-H (Figure 8A). This polymer exhibits the lowest LUMO value of −4.93 eV and demonstrates high conductivity, suggesting that polymers with lower LUMO values are more likely to achieve enhanced conductivity when doped with N-DMBI-H.
- The second design rule is derived from the properties of OEG-substituted PCNI-BTI and PCNDTI-BTI, which feature relatively small monomer units and have the predicted conductivity of 3.70 S/cm and 1.59 S/cm when doped with N-DMBI-H, respectively (Figure 8B). These polymers exhibit potential for high conductivity, indicating that the increased electron density of states introduced by the OEG side chains may enhance charge transport.
- The third design rule is derived from the high predicted conductivities when doped with N-DMBI-H of OEG-substituted P(TDPP-CT2) (1.78 S/cm), TBDPPV (1.63 S/cm), TBDOPV-T (1.66 S/cm), and TBDOPV-2T (1.46 S/cm), all of which exhibit high conductivity (Figure 8C). The predictions for these polymers suggest that an extended conjugated backbone is a crucial factor in achieving higher conductivity when doped with N-DMBI-H.

4. CONCLUSIONS AND PERSPECTIVES

The goal of this study is to predict conductivity in electron-doped organic polymers through a data-driven approach. We established a database of 84 experimentally tested n-type polymers, dopable by N-DMBI-H, collected from the literature. In parallel, a fusion deep learning model combining CNN and ANN was developed and trained to analyze polymer structures and DFT-computed descriptors. Using this model, seven polymers with OEG side chains were identified as having high conductivity. Moreover, the model demonstrated high predictive accuracy in a double-blind experiment comparing experimental measurements and ML predictions.

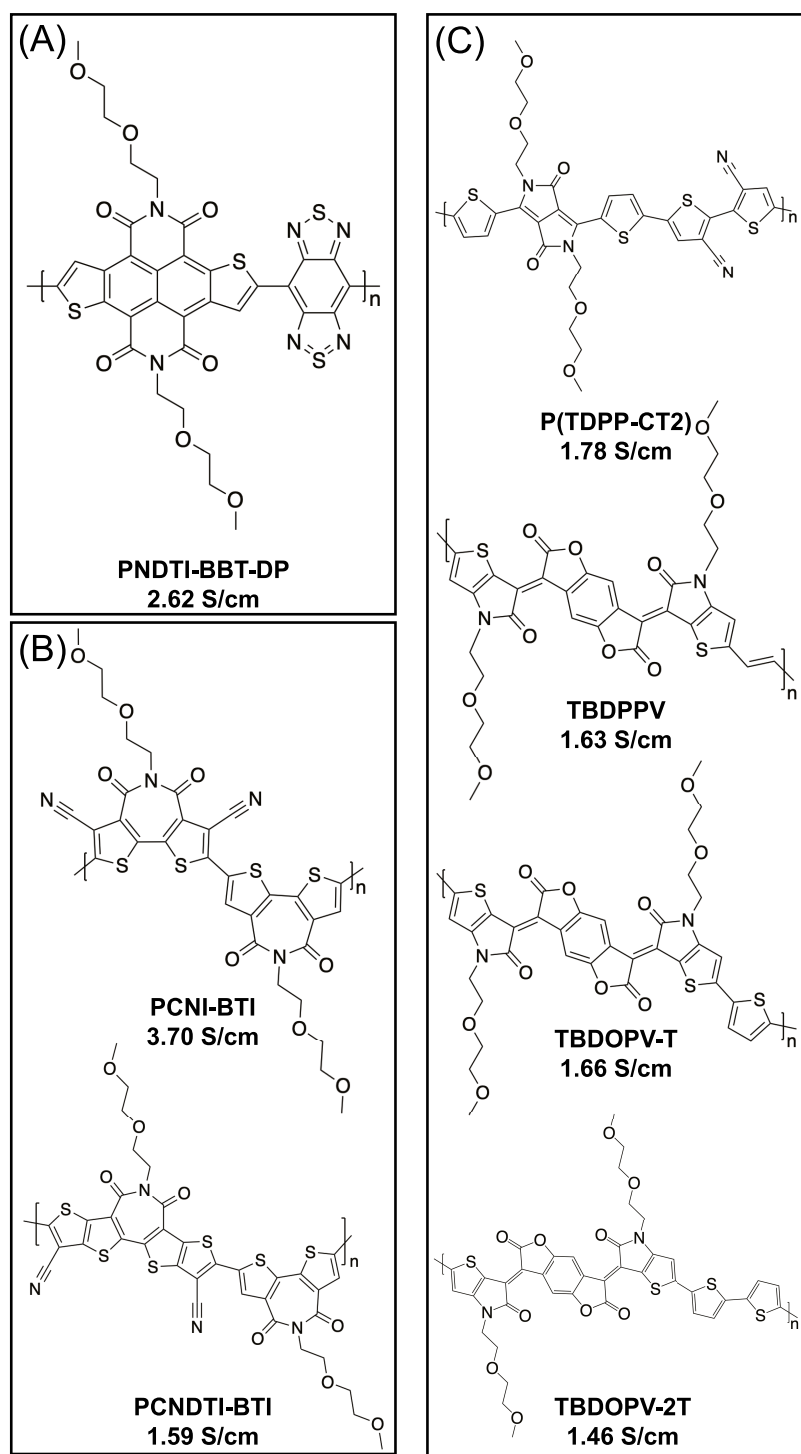


Figure 8. Structures of the seven OEG-substituted polymers with predicted N-DMBI-H doped conductivities using the fusion deep learning model.

In the double-blind experiment, the experimental group synthesized five unknown polymers, four BDPPV-type and one N2200-type, and measured their conductivity. The predicted conductivity for the N2200-type polymer, as well as for two of the BDPPV-type polymers, matched the same order of magnitude as the experimental values. This result demonstrates the model's accuracy and reliability for N2200- and BDPPV-type polymers, indicating its utility for rapidly identifying promising conductivity candidates. Nevertheless, experimental validation remains essential before making the materials-design decisions.

Our analysis of the predictions for polymers with OEG side chains revealed three main factors that influence conductivity when doped with N-DMBI-H, providing a foundational framework for designing high-conductivity n-type polymers. Specifically, lower LUMO values are associated with higher conductivity, and polymers derived from small monomers with short conjugated backbones benefit significantly from side chains that enhance electron density. In contrast, for polymers with long conjugated backbones, the side chain's impact on conductivity appears to be minimal, with emphasis placed on preserving an extended conjugated backbone with a low LUMO.

It is important to acknowledge that the database used in this work is relatively small, suggesting that the full potential of the fusion deep learning model has not yet been realized. In particular, the data set in the high-conductivity regime is limited, and cross-validation indicates that the model's predictions in this range are not fully reliable. As additional high-conductivity polymers are identified through ML-guided experimental design, we plan to incorporate reinforcement learning to enhance the model's performance in this regime.

The structural descriptor in this work was derived from the minimum-energy geometry and captures only intrinsic features of the polymer molecular structure. Additional factors, such as the relative orientation between the molecular dopant and the n-type polymer,⁷ polymer packing, and the structural flexibility of the polymer, may also play important roles in conductivity. These aspects will be investigated in future studies to provide deeper mechanistic insights into the structure–property relationships governing polymer conductivity.

This work focuses on polymer conductivity n-doped by N-DMBI-H, rather than on general dopant–polymer complex formation. As more high-quality experimental data become available for different types of n-dopants, an additional dimension incorporating these dopants will be integrated into the fusion deep learning model, enabling enhanced capabilities to search for and design highly efficient n-type dopant–polymer complexes.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c09172>.

Details of the database, parity plots, an example of sorted Coulomb matrix, experimental conductivity measurements in the double-blind test, and properties of polymers with OEG side chains (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Office of Naval Research (MURI Award 5 N00014-23-1-2001). Computations were facilitated through the use of advanced computational, storage, and networking infrastructure provided by the shared facility supported by the University of Washington Molecular Engineering Materials Center (DMR-2308979) via the Hyak supercomputer system.

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